



**TAL
TECH**

NATIONAL LEGISLATION AND e-GOVERNANCE

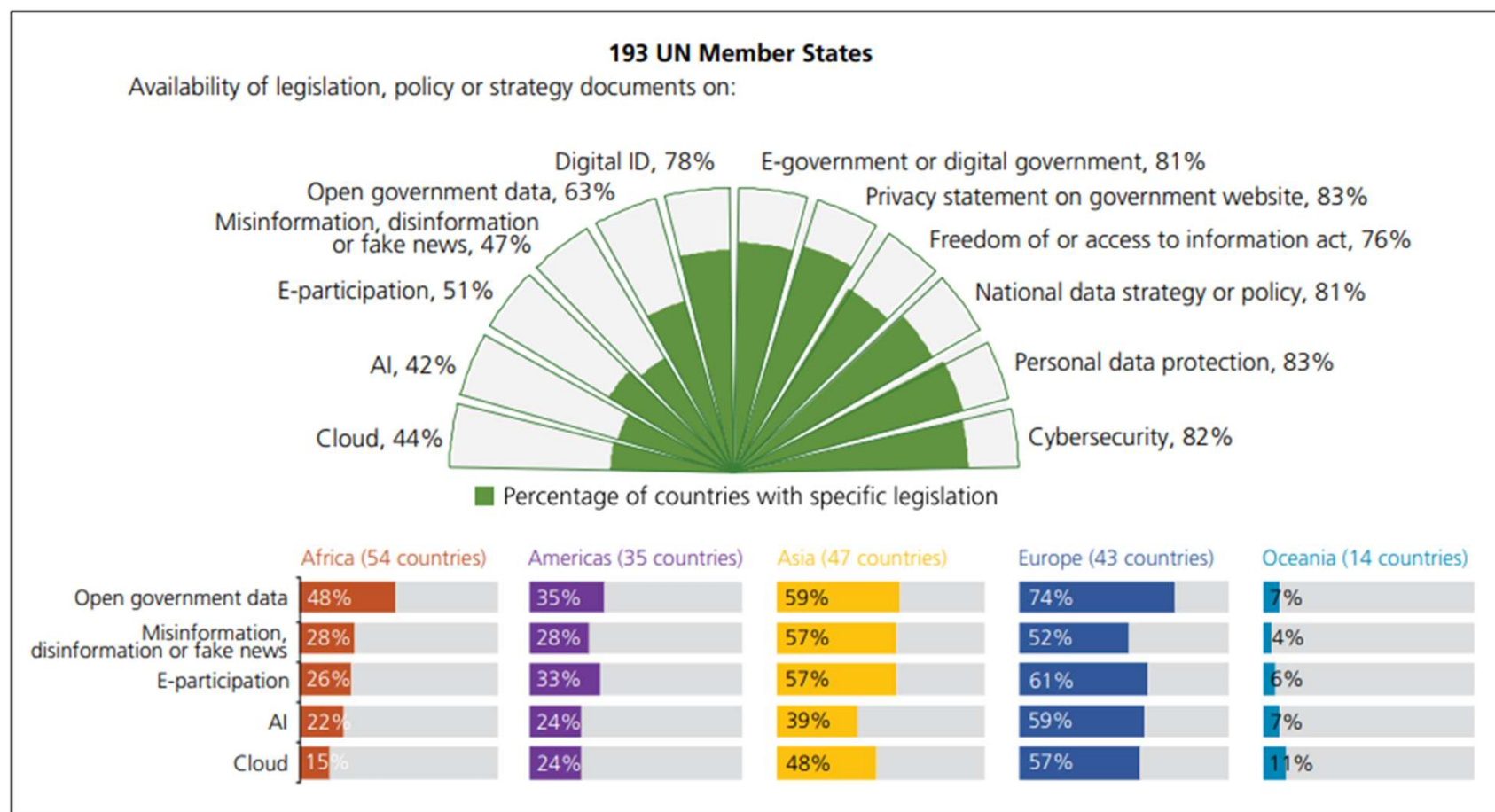
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Tallinn University of Technology

18.09.2025

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**What were your takeaways from the
last lecture?**

Figure 2.11 Percentage of countries with legislative frameworks relevant to e-government development, 2024



Source: 2024 United Nations E-Government Survey.

OUTLINE

- Organisational matters and information exchange
- Guest lecture by Dr. Rozha Kamal Ahmed „Digital Justice in Action: e-Court Systems Implementation and the Impact of Regulatory Frameworks”
- Strategic basis of e-governance at the national level
- Introduction of national legal acts
- Seminar work

ORGANISATIONAL MATTERS AND INFORMATION EXCHANGE

- **Presentation groups**

- does everybody know their group?
 - I will upload the detailed presentation guidelines to Moodle tomorrow 19.09.25.
-
- Please take time to watch an additional video lecture „The Legislative Process of Technology in Practice“ by Mr. Silver Lusti (available in Moodle under today´s lecture)

FORUM DISCUSSION - CHALLENGES

- Thank you very much to everyone who contributed by making a forum post! I found them all incredibly interesting to read!



Challenges in regulating new technologies

- Rapid tech development outpaces legislation.
- Global nature of tech demands international coordination.
- Ethical, privacy, and security concerns are hard to anticipate.
- Risk of overregulation vs. legal gaps.
- Policymakers often lack clarity on which technologies to regulate.
- Regulation is resource-intensive and requires deep understanding.

FORUM DISCUSSION - SOLUTIONS

- Strengthen regulators' skills and tech awareness.
- Use existing laws with adaptive interpretation.
- Employ AI to assist in drafting and updating laws.
- Streamline legislative processes (e.g., directives over regulations).
- Track investment trends to identify high-impact technologies.
- Establish tech-watch mechanisms within regulatory bodies.
- Collaborate with private sector and civil society for insights.

FORUM DISCUSSION – NEW TOPIC!

- Balancing accountability and innovation
- How can we ensure accountability for harmful or fake AI content, while still enabling AI to be used for positive purposes such as creativity, learning, and healthcare?



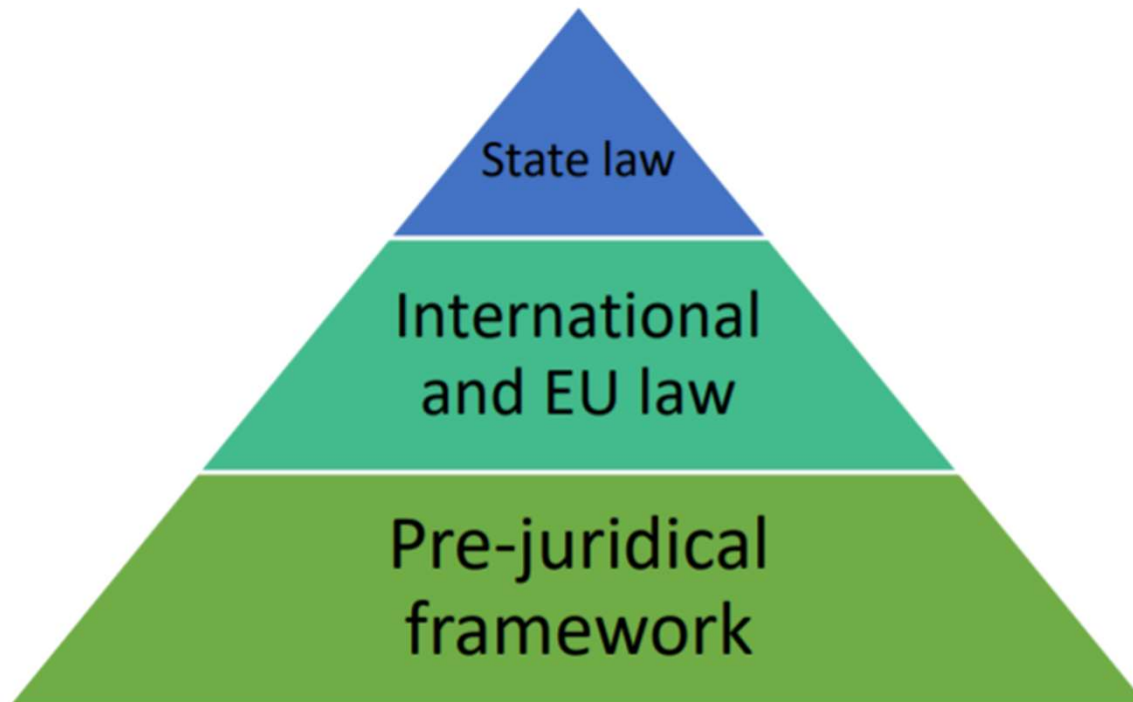
Digital Justice in Action: e-Court Systems Implementation and the Impact of Regulatory Frameworks

Dr. Rozha Kamal Ahmed

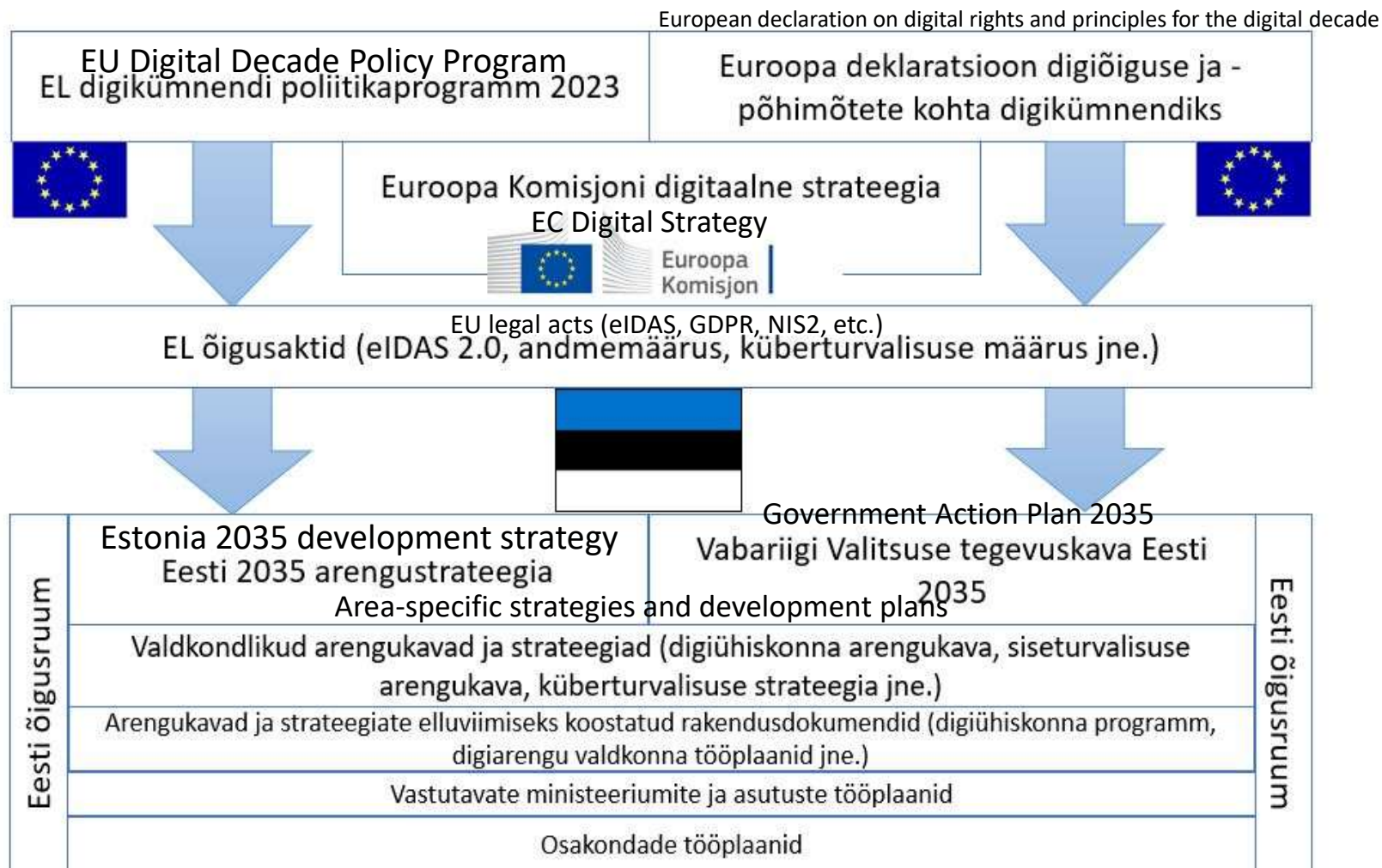
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Introduction to the national legal acts

LEGAL AND REGULATORY FRAMEWORK

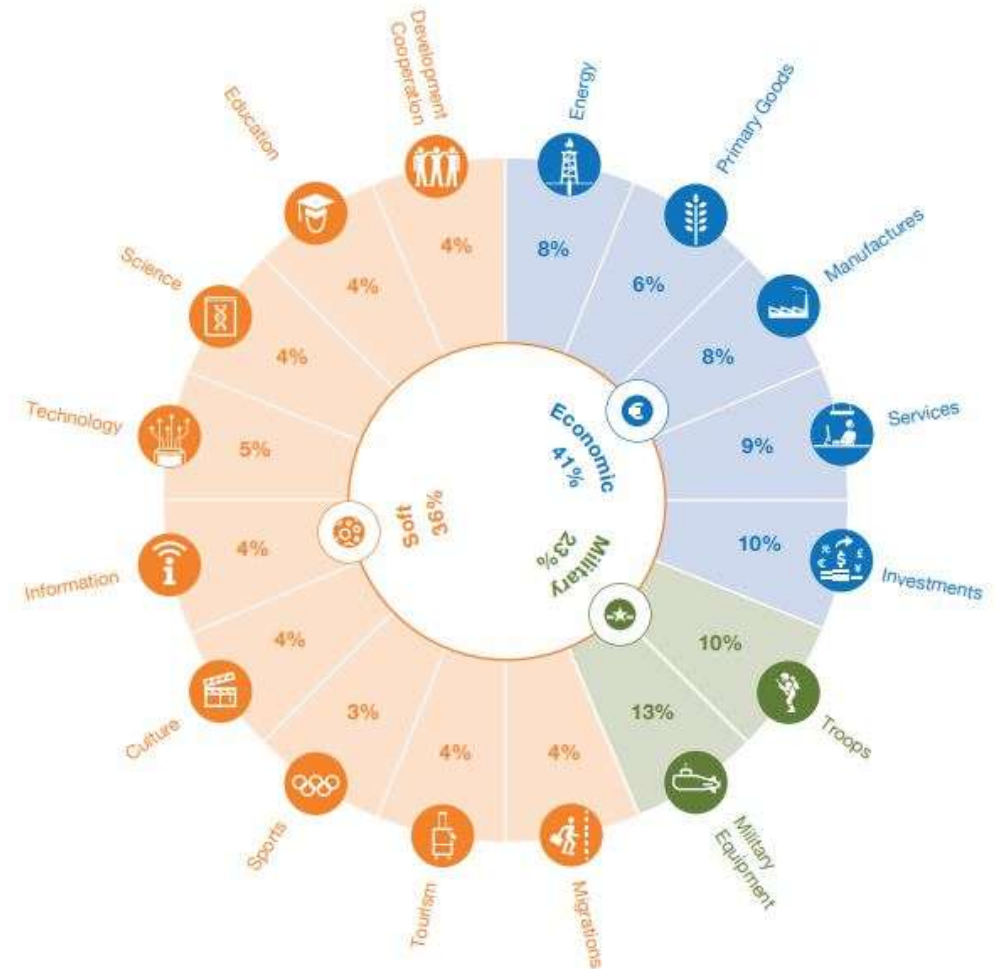


STRATEGIC BASIS OF E-GOVERNANCE



GLOBAL PRESENCE INDEX (ELCANO INDEX)

The Index shows country's position related to global visibility/presence in economic, military and other areas (tourism, culture, technology, education, science, etc.)

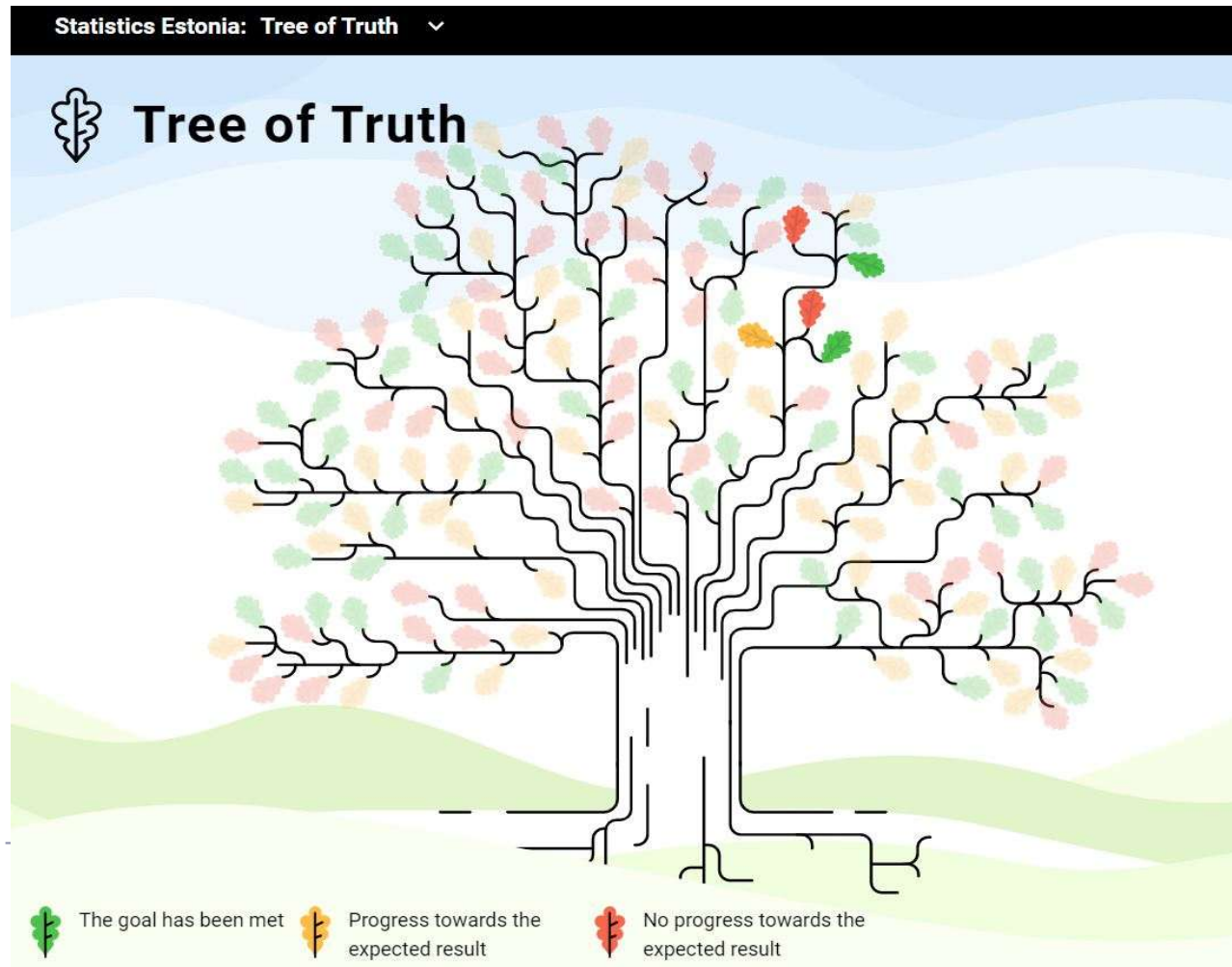


THE VISION IS...

*„Estonia is an **innovative country** that **values the creation and use of knowledge**, where social life is organized with the help of new, people-centered, and efficient technologies. The legal and tax environment and the organization of governance foster social cohesion, the adoption of new solutions, innovation, and flexible public administration. Public **services function in the background** and **are predictable**, and **the data space is protected**. Organization of the state and people's participation therein is both trend-setting and exemplary for other countries.“*

TREE OF TRUTH

<https://tamm.stat.ee/kategooriad/eesti-areng?valdkond=riigivalitsemine&lang=en>



ESTONIAN DIGITAL AGENDA 2030

- It was adopted by the government in October 2021.
- VISION: Estonia, empowered by digitalization
- Overall funding planned for 10 years is 1,2 billion euros

Principles:

- protection and promotion of the fundamental rights of people.
- Preserving the Estonian language and culture
- Reliability and prevention
- Technology-neutrality
- Building a digital society together
- Innovation
- Climate and environmental friendliness



Development agenda
of the field
2021

Based on the vision, the goal for the next decade in terms of the development of digital society is to increase Estonia's digital power: **digital government guarantees the best experience, high-speed Internet is available to all those who request it in Estonia and our cyberspace is safe and reliable.**

HOW TO MEASURE?

$$\left(\begin{array}{l} \text{satisfaction} \\ \text{of private} \\ \text{persons with} \\ \text{public digital} \\ \text{services} \end{array} + \begin{array}{l} \text{satisfaction of} \\ \text{entrepreneurs} \\ \text{with public} \\ \text{digital services} \end{array} + \begin{array}{l} \text{avail-} \\ \text{ability} \\ \text{of high-} \\ \text{speed} \\ \text{Internet} \end{array} + \begin{array}{l} \text{resilience} \\ \text{and trustwor-} \\ \text{thiness of} \\ \text{cyberspace} \end{array} \right) / 4 = X$$

Initial rate 67,5 – expected rate by 2030 94

PROMISES

1. Leaps in the development of digital government in Estonia

Promises: the greatest results by 2030 at the latest	Line of action
In Estonia, public services reach you just when you need them and you can complete all procedures at once.	Switch to life and business event based and proactive services
All procedures concerning the public sector can be carried out using chatbots in Estonia	AI-powered government
Another leap has been taken in increasing efficiency in the public sector by means of AI	AI-powered government
The fundamental rights of people are protected in digital government and people and companies have control over their data and the opportunity to actually share them	Human-centric digital government
Estonia is the greenest digital government in the world	Green digital government

DIRECTIONS

2. Directions which enable us to take a leap in development and ensure the sustainability of digital government

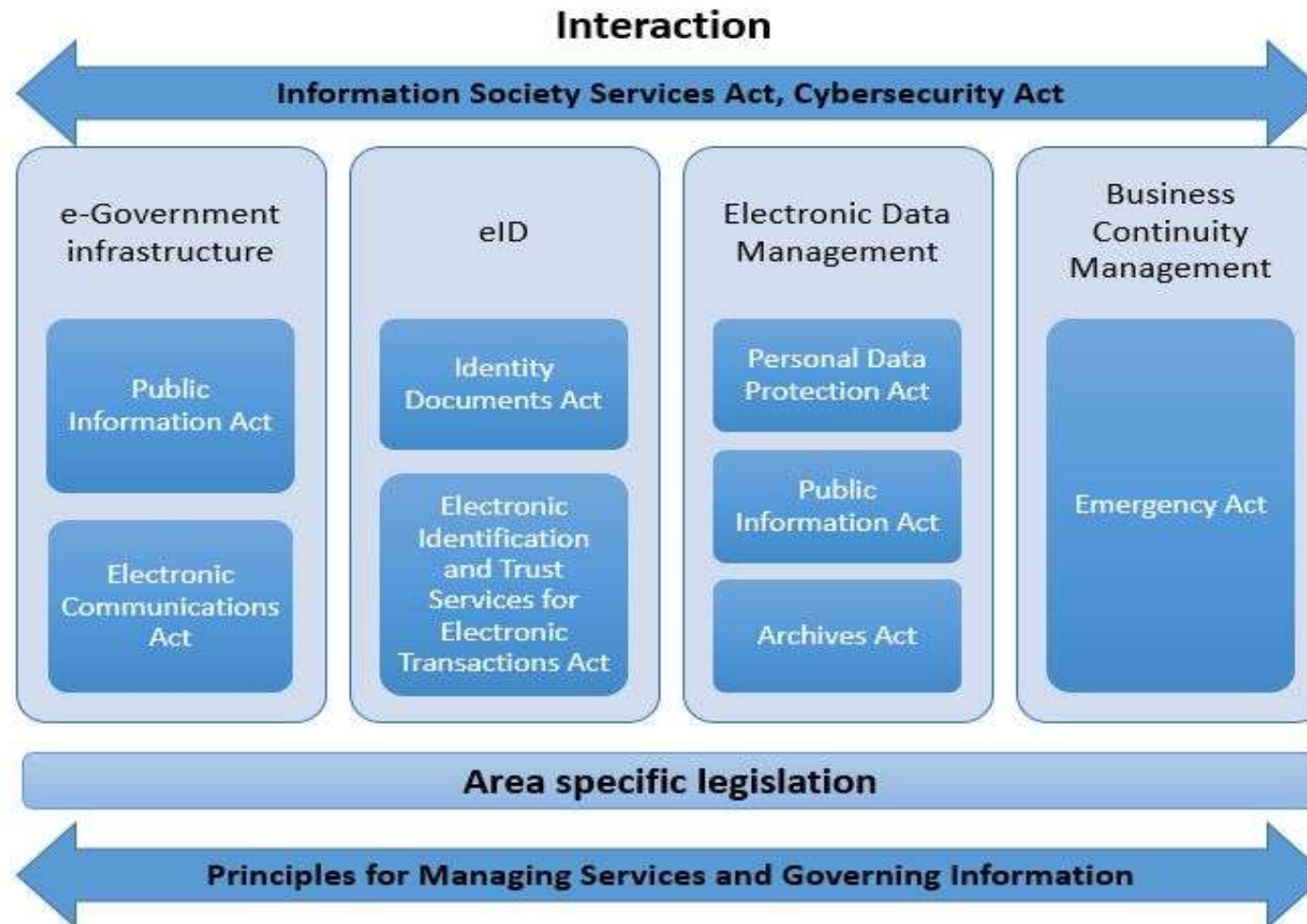
Promises: the greatest results by 2030 at the latest	Line of action
All public services are human-centric and designed, managed and measured in a uniform manner	Introduction of the management and user-centricity of public services
All the decisions of the state are made on the basis of high-quality data	Data-driven governance and reuse of data
The common solutions ⁹ of digital government are futureproof	Future-proof digital government platforms
Digital government is cloud-native and standard services are of high quality	Centrally provided standard IT services
Estonia is a pathfinder and an eager experimenter	Systematic experimentation with new ways
The private sector makes a significant contribution to the innovation of digital government and solutions can be taken to the world	Open innovation
We have the knowledge, skills and funds required for bold and large-scale digital changes	Empowering digital change in public sector
There are more and more cross-border services.	Targeted international cooperation

TYPES OF NATIONAL LEGAL ACTS

- Laws and regulations (Government or minister)



CASE OF ESTONIA



INFORMATION SOCIETY SERVICES ACT

- Provides for the requirements for information society service providers, the organization of supervision, and organizational liability.
- **“Information society services”** are services provided in the form of economic or professional activities at the direct request of a recipient of the services, **without the parties being simultaneously present at the same location**, and such services involve the processing, storage, or transmission of information by electronic means intended for the digital processing and storage of data. Information society services must be entirely transmitted, conveyed, and received by electronic means of communication.
- **Services provided by means of fax or telephone call and television or radio services are not information society services.**
- Supervision is provided by the Consumer Protection and Technical Regulatory Authority.

CYBERSECURITY ACT

- Provides requirements for the maintenance of network and information systems essential for the functioning of society, including network and information systems of the public sector, liability and supervision, as well as bases for the prevention and resolution of cyber incidents.
- **Does not apply to:**
 - The processing of state secret and classified information;
 - Military cooperation;
 - Digital service providers with less than 50 employees with a total annual turnover of less than 10 million euros.
- **Obligations arising from the law**
 - Obligation to apply security measures for preventing and resolving cyber incidents.
 - Obligation to notify a cyber incident to the Estonian Information System Authority.

E-GOVERNANCE INFRASTRUCTURE

- Basic infrastructure for service provision and state data exchange architecture.
- Legal Acts
 - Electronic Communications Act

„The aim of the act is to create the necessary conditions for the development of electronic communications to promote the development of electronic communications networks and electronic communications services without giving preference to specific technologies and to ensure the protection of the interests of users of electronic communications services by promoting free competition and the purposeful and just planning, allocation and use of radio frequencies and numbering.“

- Public Information Act

„The aim is to ensure that the public and every person has the opportunity to access information intended for public use.“

DATABASES

- A database is a **structured body of data** processed within an information system of the state, local government, or other person in public law or person in private law performing public duties, which is established and used for the performance of functions provided in a Public Information Act.
- Data collection in the database shall be based on the one-request-only principle.
- **The state information system** consists of databases that are connected to the data exchange layer of the information system and registered in the administration system of the state information system and of the systems supporting the maintenance of the databases.

ESTABLISHMENT OF DATABASES

- The establishment of separate databases to collect the same data is prohibited.
- Before the establishment of a database or changing the composition of the data collected in a database, introducing a database, or terminating a database, the technical documentation of the database shall be approved by:
 - the Estonian Information System's Authority
 - the Data Protection Inspectorate
 - the Statistics Estonia.
- Chief and authorized processors of databases.
- Statute of the database.

STATE INFORMATION SYSTEM MANAGEMENT SYSTEM



RIIGI INFOSÜSTEEMI
HALDUSSÜSTEEM

[Avaleht](#)

[RIHA kataloog](#)

[RIHA varamu](#)

[Abikeskus](#)

Otsi

Ülevaade riigi infosüsteemist

Riigi infosüsteemi haldussüsteemist RIHA leiad riigi infosüsteemide ja andmete kirjeldused

500

Aktiivset asutust ja ettevõtet

üle 1300

registreeritud infosüsteemi ja andmekogu



RIHA kataloogi sirvimine

Kõik RIHAsse kirja pandud infosüsteemid on leitavad avalikust RIHA kataloogist. Kataloog annab infot, kui palju erinevaid süsteeme Eesti riigi infosüsteemi üldlõdse moodustavad.

Infosüsteemi sirvimine RIHA-s:

- annab teadmise erinevatest infosüsteemidest riigis, nende eesmärgist ja andmetest
- hõlbustab infosüsteemi eest vastutavate inimeste leidmist
- annab infosüsteemi omanikule teadmise, kust leida talle vajalikku andmekomplekti

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Infosüsteemi haldamine

Selleks, et teised leiaksid sinu infosüsteemi, tuleb see esmalt kirja panna. Olgu tegemist alles loodava või juba aastaid kasutuselolevaga, siis kirjelda see RIHAs.

Infosüsteemi kirjeldamine RIHA-s annab:

- teistele asutustele ja arendajatele teada, mis andmed sul on ja kuidas neid kasutada saab
- teadmise, kas infosüsteem vastab riigi IT nõuetele
- sulle endale kindluse, et oled täitnud tublisti seadust

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X-teega liitumine

Riigi infosüsteemidega turvaliseks ja koosvõimeliseks andmevahetuseks on võimalik kasutada X-tee andmevahetuskihti.

X-teega liitumine võimaldab:

- keskenduda infosüsteemi funktsionaalsusele ning jätta andmevahetuse turvalisus X-tee hooleks
- kasutada riigi infosüsteemide teenuseid oma äriprotsesside optimeerimiseks
- pakkuda teistele turvalist ligipääsu enda andmetele

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eID

- **Identity Documents Act**

Establishes identity document requirements and regulates the issue of identity documents to Estonian citizens and aliens by the Republic of Estonia.

- **Electronic Identification and Trust Services for Electronic Transactions Act**

Regulates electronic identification and trust services for electronic transactions and organization of state supervision to the extent that is not regulated at the EU level (for example, eIDAS regulation).

- **Population Register Act**

Legal basis for documenting the population and personal identification code.

IDENTITY CONCEPT IN ESTONIA

- Identity is based on a unique identifier (personal identification code, 11 digits)
- **xYYMMDDyyyz**
 - x- century and sex
 - YYMMDD – date of birth
 - yyy – localization and sequence number on the registration day
 - z- control number
- All identities connected to this one unique number.

ID-CARD

- Issued by Estonian Police and Border Guard Board
- 5 cards in the same family (ID card, residence permit card, digital identity card, e-residency card, diplomatic identity card)
- Mandatory identity document with eID functionality
- Enables authentication, signing, and encryption
- 2 pairs of PKI keys with corresponding X.509 certificate
- Valid for up to five years.



MOBILE-ID

- Issued by local mobile operators
- Special SIM card with PKI eID chip
 - SIM Toolkit application
 - SMS service for data exchange
- 2 pairs of keys with corresponding X.509 certificate
 - Private keys in a secure module on SIM
 - Certificate stored in the public repository only
- Central M-ID Service instead of middleware
- Enables authentication and signing
- Validity for up to 5 years.



The logo for TAL TECH, with 'TAL' stacked above 'TECH' in a bold, white, sans-serif font. The background of the slide features a blue-to-purple gradient at the top and a solid purple area on the right where the logo is placed.

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**Does the personal identification code
should be random or it is ok to have
some pieces of personal data included?**

PERSONAL DATA PROTECTION

- Personal Data Protection Act and main data processing principles:

1) **legality and fairness** – personal data are processed legally and fairly;

2) **purposefulness** – personal data are collected for specified, explicit and legitimate purposes and they shall not be processed in any manner which is incompatible with these purposes;

3) **quality** – personal data must be adequate and appropriate and must not be excessive given the purposes of the data processing;

4) **accuracy** – personal data must be accurate and, if necessary, kept up to date; reasonable measures are taken to ensure that any personal data which are inaccurate with respect to the purpose of data processing shall be erased or rectified without delay;

5) **retention** – personal data are retained in the format which enables to identify the data subject only until this is necessary for achievement of the purpose for which the personal data is processed;

6) **security** – personal data are processed in a manner that ensures appropriate security thereof, including protection against unauthorised or unlawful processing and against accidental loss, destruction or damage, by means of implementing appropriate technical or organizational measures.

SOLUTIONS THAT SUPPORT DATA PROTECTION

- **Data tracker** - provides the citizen with a clear overview of the operations performed with their data. A complete overview is displayed on the state portal eesti.ee.
- **Consent service** - allows a person to give permission to the state to share their personal data with a certain service provider.

BUSINESS CONTINUITY MANAGEMENT

- Plan B for mature e-governance.
- How to ensure electronic service provision in emergency cases (for example cyber-attacks).
- Important to be aware of dependencies.

E-GOVERNANCE RELATED NATIONAL LEGISLATION

Estonian legislation examples:

- Information Society Services Act (from 2004, last change 14.07.2024);
- Cybersecurity Act (from 2018, last change 01.07.2024);
- Electronic Identification and Trust Services for Electronic Transactions Act (2016, last change 13.03.2023);
- Personal Data Protection Act (2018, last change 01.11.2023);
- Public Information Act (2001, last change 01.05.2023);
- Population Registry Act (2019);
- Identity Documents Act (1999);
- Electronic Communications Act (2004);
- Competition Act (2001);
- Copyright Act (1992);
- Emergency Act (2017);
- Penal Code (from 2001);



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SEMINAR WORK



- We will be conducting a **seminar on national e-governance legislation**

The seminar will be structured as follows:

- **Individual Research:** Each student will have 10 minutes to use ChatGPT (or any other similar AI tool) to research their home country's e-governance legislation. Focus on key aspects such as the scope of the legislation, its objectives, and any significant challenges or successes in implementation, etc.
- Participants will form groups (5-7 people in a group).
- **Group Discussion:** Spend the next 15 minutes discussing your findings and comparing the similarities and differences across the various countries. Reflect on the implications of these variations for e-governance practices and development.
- **Summarizing the Findings:** Finally, each team will have 5 minutes to present their key findings and comparisons to the class. Highlight the most interesting or noteworthy aspects of the legislation you examined.

The TAL TECH logo is positioned in the upper right corner. It features the words 'TAL' and 'TECH' stacked vertically in a bold, white, sans-serif font. The background behind the logo is a horizontal gradient transitioning from light blue on the left to a vibrant magenta on the right. A decorative magenta shape, resembling a stylized 'L' or a series of steps, is located on the left side of the slide, extending from the top to the bottom.

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THANK YOU VERY MUCH!

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